having a steam-tight job. Many of our readers have, no doubt, frequently seen seams made tight and then break out in spots a little later on. These leaks are caught only to break out in another place. The diagonal pitch in a case of this kind is generally too great.

To Ascertain the Lap.

The amount of lap is varied according to the ideas of those who handle the work. A short lap is desired, when the seam is exposed to flames or heat, so as to prevent the sheets cracking from the rivet holes to the calking edge. The water being unable to reach the sheet and rivet head directly, causes the material at this point to get hotter, resulting in cracks. Therefore, as short a lap as possible is used when the seam is directly exposed to the fire and heat. Some boiler makers have resorted to counter-sinking the rivet holes, and are driving an oval counter-sunk rivet, as shown in Fig. 6. The rule generally used is to make the lap 1½ times the diameter of the rivet hole. This is sometimes varied by taking 1½ times the diameter of the rivet, which, of course, gives a slightly smaller lap, as the diameter of the rivet is 1/16 inch less than the diameter of the hole.

(To be Continued.)

Locomotive Flue Welding.*

Flue work in our railroad shops is one of the important classes of work, as much so as locomotive frames, rod work, motion work or the different classes of locomotive work which we take up from year to year. Our association has taken it up right along and some improvement has been shown at the conventions when the subject was considered.

IN THE METHOD OF WELDING, SHOULDERING AND EXPANDING FLUES.

Our first aim should be to get a solid weld. This is the most important of all operations. In regard to shouldering or expanding, there is no question, as these operations are worked from the solid end and need no special care only to get to the sizes.

The welds, on the other hand, must not be defective. If they are defective and show a leak after being put in the boiler, the flues must be taken out and all work done on that flue gone over again, which doubles the cost on every flue that is defective, not counting the delay of holding the engine out of service, which is quite an item sometimes on that account.

The second point is the output. To get a good output, a fair day's work is what we are duty bound to do, in order to hold a good record and satisfy our company and keep our shop up to or ahead of any other shop. We should get good average day's work every day.

We do not consider these grandstand plays made on exhibition and made to show what can be done by the different kinds of tools and furnaces to be a fair average, for the work done on those trials is not up to the standard of what we must do right along. We have seen some of those tests made which did not show a good quality of work done, but showed great speed, or how fast a certain furnace would heat and how fast a certain machine could weld flues.

THE DIFFERENT METHODS OF PREPARING FLUES AND WELDING SAME.

Some shops, after the flue is cut off, scarf the flue and safe end by machine back about ½ inch to a thin edge, and then weld. Other shops scarf under their welding machine and then weld both the safe end and flue. This practice has been followed for years.

While a great many shops of to-day have abandoned the idea of scarfing either the flue or safe end as an operation, yet when the flue and safe end are cut off, they are cut by a disk or round cutter, beveled on one side and straight on the opposite side, so that when the flue or safe end is cut it leaves the end beveled ready for welding. This does away with the expense of scarfing. When the flues and safe ends are cut this way it leaves them scarfed at no cost and there is no waste of metal as there are no cuttings.

In welding the flues and safe ends that are cut in this way, there is a mandrel, that is tapered at one end, placed alongside of the welding furnace. The flue welder's helper puts the flues right from the cutter into the furnace, gets the end hot, shoves it up on the mandrel, which opens out the end. He has his safe end laying in the furnace, returns the flue to the furnace and pulls the safe end back into the flue and continues to heat for the welder. The furnace has two heating holes. The helper keeps the flues ready for the welder as fast as he welds and shoulders. On an ordinary safe end he welds and shoulders at one heat.

The method of cutting the flues this way leaves a short bevel or scarf, which, when put together, makes a nice solid weld from the fact of giving you plenty of metal right through the weld, and you must work the metal in and down to size. This makes a good job.

One practice we want to call your attention to is the cross-bar at back end of furnace that is intended to jump or upset the flue against when it is hot, before taking same to the hammer or welding machine to be welded. This is important, as it puts the safe end back against the flue and upsets it where it is welded, and there is where you need the material.

We have found some flue welders who seemed to be in a hurry and would not take the time to jump the flue when hot against this bar. We do not think this is good practice, as we believe that flue work should not be slighted.

PRACTICAL SUGGESTIONS.

For a flue shop, where it can be done, it seems to be the old custom on flue work to have the rattler or cleaner in some out-of-the-way place, and the cutting-off machine in another place not very convenient to the welder. This causes the flue to be carried from the truck to the cleaner or rattler and from the rattler to the cutting machine, and after cutting carried again to the welder and very likely from the welder back to the cutting machine again to be cut to length. All of this carrying is costly and unnecessary if it can be avoided.

The new plan which is in practice and is found to be a good plan, is to place the flue shop in line with the rattler, which is at one end of the shop and high enough above the ground so that the flues when rattled can be dropped out from under the rattler and roll to the flue cutter on inclined rails. When the cutter cuts them and drops them down, they then roll to the welder. They are welded and shouldered; then they are dropped down and roll to another cutter, which cuts them to length and then they roll to the expander, if you expand them. All is done and not one was carried or costs a cent for portage through the flue shop, being handled only by the operator. The flue is not even turned around, but goes straight through to a finish, there being two cutting machines, one at each end of flue and one placed close to rattler, the other placed after the flues leave the welder.

No person touches the flue after the rattler is opened, so far as portage is concerned, until the flue is carried to the engine, it being handled only in cutting and welding.

FUEL FOR HEATING FLUES USED AT PRESENT.

Hard coke has been the old reliable, but it has seen its day as it costs too much to handle it, and we lose too much time in cleaning the furnace and wheeling out the ashes, and the

*A paper presented before the convention of the Railway Master Blacksmiths' Association.

Answer—We do not use the entire steel flue; just use the steel ends.

Mr. A. N. Lucas—We have considerable pitting with the iron flue. The last few years the iron flues pit considerably. Tests show that there is some steel in them; they are semi-steel.

In reply to questions Mr. Lucas stated that he allowed a scant quarter inch for the bead and used a prosser expander with a shoulder on it for new work.

Mr. Green—I believe the life of a flue depends largely on the thickness of the sheet. I have found that engines with 5/8 flue sheets would run possibly four months, and then begin to burn the beads off, and in the same district a 1/2-inch sheet would run longer. It is not so much in setting the flues because most everybody uses a prosser expander.

Mr. D. A. Lucas—I have had considerable experience with flues and have treated them different ways. Our present method is filing the flue hole through a radius of 1/16 of an inch on each side and applying a copper ferrule flush with the sheet on the fire side. We use an inch long shim.

I would like to ask whether the flue leaks between the flue and the copper or between the copper and the sheet? I made an examination once and found that it leaked between the copper and the flue. With a long shim the copper shim is rolled into the sheet and the flue swedged 1¼ back, and it makes a nice dry fit. When the cold air hits the flue and strikes it in the sheet proper, I did not experience the weeping of the flues that I did with the narrow shim.

When they put the engine into the round house to give it a final touching up, I have the flues rolled lightly and calked, and we have practically stopped the flue trouble. We are in a bad water district where the flues won't run over a year on account of pitting. We are using the Tyler tube for our iron tube, and we have had several test sets of steel tubes with good results.

Mr. A. N. Lucas—A great many times in doing round house work, the flues leak badly and quite often the boiler maker gets instructions to go after them hard. The harder they go after them the worse the results. They crack the bridges and pull the flue sheets out of shape. You cannot work a flue too lightly; just work it in the sheet and do not work the sheet itself.

Mr. Lewis—I went to the Lehigh Valley last January and found conditions similar to what Mr. Hennessey spoke about on the New York Central, and I went into the question of the tools the men were using. I found them all using the roller expander, and they were rolling and rolling and rolling. They kept a record of what engines they had worked on and everything had been rolled.

I reported to my superior officers that the practice would have to stop if they wanted to maintain the life of the flue. I said, "Take the rollers away from the men and give them a prosser tool, and see that they are furnished with the proper beading tools." After discontinuing the use of roller expanders in round house work, the reports concerning engine failures have been very encouraging.

I think the proper tool to be used to maintain the life of the flues is a prosser tool, one with the shoulder on and one with the shoulder off.

I might say that I have changed the method of applying flues. I have put into practice the same method as Mr. Lucas uses, with the exception that he does not roll the flues. I believe that when we first apply the flues we ought to give them a light rolling after the prosser has done its work. Old flues, which had been rolled thin, occasionally burst when prossered, but new ones never did.

Mr. Lester reported a similar method of applying flues in vogue on the Erie.

Mr. Plowman—We have eighteen or twenty extremely wide fire-boxes, and the greatest trouble is in the flue sheet bulging at the bottom. We put in a set of flues and they run from four to seven or eight months when the flue sheet will bulge as much as 1⅝ inches at the bottom. We adopted the plan of leaving out the open space of flues and bracing them 1¼ inches. We find that the life of our flues is from seven to eight months longer with our flue sheets braced.

Mr. T. J. Morrison—There has been a great deal said about flue setting, but there is as much in taking care of the flues after they are set in the boiler as there is in the setting. We take them out and clean the flue holes out—clean out the dirt which accumulates. We drive them tight and roll the flues lightly after we put them in; then I have them laid over with a hammer, and when the man is laying them over I have them rolled in the front end before I have them expanded in the back end. And then we bead them.

In regard to the engine after it goes out. There is a great deal in the care that the fireman will give an engine on the road. Some firemen and engineers will get better service out of their flues than others, because they keep their fire level. If more care were taken of the engines at the terminals and more care taken of the boilers at washouts it would be better. We have been able to dispense with one boiler maker on repairs. We blow off the boiler through the pipes, wash with hot water and fill the boiler again with hot water. The flue sheet and fire-box is never cold, and we do not have much trouble with flues or cracked side sheets. I thought at one time that there was nothing like rollers, but we now use a prosser expander. When a flue is rolled too often it stretches and becomes thin, and you do not have enough material to work on to keep it tight. Since we have been using the prosser we get from eight to nine months out of the lower half, then we take out about 100 or 125 and renew them on the bottom.

Mr. Flavin—It is one of the greatest mistakes to roll a flue after it is prossered. Twenty years ago, at the International Railway Exhibit, I had charge of the flue demonstation of the C. & A. road. The first flue was rolled and beaded; the next was prossered and beaded, the next was prossered and rolled and beaded, the last was prossered, beaded and then rolled. We found in the case where the flue was prossered and beaded the joint on the fire side was perfect, but there was no joint between the edges of the sheet. In the case of the flue that was prossered and then rolled we found that the entire labor of prossering the flue was wasted. The flue that was rolled and then beaded showed a very pronounced joint on the fire side, but not nearly so pronounced a joint on the water side. The taper end of the flue on the fire side will roll more than on the inside. After that the practice was to roll the flues lightly, then prosser and bead. Another objection to the roller is that the copper ferrule is put in there on account of its being fibrous, but I have visited shops where the roller is used to such an extent that the copper ferrule is entirely useless. The roller should never be used hard, and after the flues are once in service the roller should never be used on them unless it is absolutely necessary.

There should never be a given limit to the length of the flue to be turned over because the flue bead has one more thing to do when it stops a leak and that is to brace the sheet.

There is no excuse to-day for a plugged flue. If it is bursted it should be removed, which is a very simple matter. I believe the ferrule should be exactly the same thickness as the sheet.

Another thing to be considered is the size of the nozzle. With a self-cleaning front end it is wrong to apply a set of nozzles so small that the entire draft of the fire-box goes in every time the fireman opens the door. We will never solve the flue proposition until we solve the nozzle proposition.

President Smythe—You have noticed in many cases in beading a flue that it would lap a burr on the inside; you would not